

TYPED CONTROL, INFORMATION, AND SADDLE ENVELOPES FOR KINETIC COTANGENT PHASES

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ABSTRACT. Repeated exact observations and collision faces generate a compatible stratified current tower. Filtering uses unnormalized coarea currents together with a log-evidence ledger, so rare regular observations are retained without a false uniform denominator bound. Quenched coarea coefficients and posterior cluster estimates reduce only the strategy-reachable chaotic belief family to the resolved kinetic state.

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1. ROUND-EIGHT RECONSTRUCTION: EVIDENCE-LIFTED STRATIFIED FILTERING AND REACHABLE-STATE REDUCTION

Exact observations are retained. Because their coarea normalizers may be arbitrarily small, the filtering state carries the unnormalized current and its evidence separately. Repeated observation and collision intersections are represented by a closed tower of compatible currents of every reached codimension.

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1.1. The stratified current tower. For a chronological cell let $\mathfrak{S}$ be the finite family generated by collision faces and the level sets of the observation maps used up to the current decision time. A state is a current chain

$$\mathbf{J} = (J_S)_{S \in \mathfrak{S}}$$

with J_S a finite measure on the stratum S and incidence compatibility

$$\partial J_S = \sum_{S' \prec S} \epsilon(S, S') J_{S'}.$$

Set

$$\|\mathbf{J}\|_{\text{str}} = \sum_{S \in \mathfrak{S}} \omega^{\text{codim } S} (\|J_S\|_{\mathcal{M}_w} + \|T_S J_S\|_{\mathcal{M}_w}).$$

Lemma 1.1 (Closed stratified trace complex). *The B2 transport, reflection, contact creation, and coarea restriction operators act triangularly on the completion of compatible current chains. The boundary operator squares to zero, and every new observation of rank r raises codimension by r while preserving the incidence identities.*

Proof. Start with smooth differential forms on every transverse stratum. Stokes' formula gives the incidence relation and $\partial^2 = 0$. B2 transport maps a stratum to its flow image; reflection and contact maps preserve incidence orientation. Coarea restriction is intersection with the level-set current, and transverse intersection commutes with boundary up to the standard signed incidence rule. The weighted triangular norm absorbs the finite number of new lower strata. Passing to the graph completion proves the assertion. $\square$

1.2. Evidence-lifted Bayes updates. For an observation O^u and an unnormalized predicted current $\bar{\Pi}^-$, define

$$\bar{\Pi}^y = \delta_{O^u=y} \bar{\Pi}^-$$

by coarea intersection and let

$$Z(y) = \bar{\Pi}^y(1).$$

The normalized posterior is $\Pi^y = \bar{\Pi}^y / Z(y)$ when $Z(y) > 0$, but the state used by the dynamic program is

$$(\bar{\Pi}^y, \ell^y), \quad \ell^y = -\log Z(y).$$

Lemma 1.2 (Bayes update without a uniform denominator). *The map $(\bar{\Pi}^-, y) \mapsto (\bar{\Pi}^y, \ell^y)$ is continuous in the projective stratified topology on every finite evidence slice $\ell^y \leq L$. Values with small $Z(y)$ are retained with their exact evidence cost; no uniform positive lower bound for $Z(y)$ is asserted.*

Proof. On a transverse observation chart, coarea intersection is continuous in the stratified graph norm. Projective normalization is continuous when $Z(y) \geq e^{-L}$. If $Z(y)$ is smaller, the unnormalized current still varies continuously and the diverging log evidence is an explicit state coordinate rather than a hidden norm blowup. $\square$

Theorem 1.3 (Feller evidence-filter kernel). *Assume the observation maps are uniformly transverse on each declared chart and their critical strata are included explicitly in $\mathfrak{S}$. Prediction by the B2 graph semigroup followed by the evidence-lifted update defines a Feller kernel on the compactly contained evidence state space.*

Proof. Prediction is Feller by the B2 graph semigroup. On each transverse chart apply Lemma 1.2. The critical set is not identified with the billiard grazing set; it is a separate finite current stratum and is propagated by Lemma 1.1. Exponential evidence containment removes the large- ℓ tail uniformly, so the finite chart pieces paste to a Feller kernel. $\square$

1.3. Quenched coarea coefficients. A noiseless observation inserts the Fourier representation of its level-set current. The observation frequency is therefore an additional continuous frequency in the B2/B1 coefficient theorem.

Theorem 1.4 (Uniform coarea local-limit ratio). *On every strategy-reachable compact source and evidence set, numerator and denominator coarea coefficients have one common exact saddle and satisfy uniform mixed local-limit expansions. Their ratio converges to the conditional block pressure, while the finite log-evidence term is retained explicitly.*

Proof. Use Fourier inversion for the collision/lattice constraints together with the observation delta. Transversality gives a full-rank observation block; the B1 good-block regeneration gives high-frequency decay, and B2 source sewing gives uniform bounded-source derivatives. Major-arc Gaussian coefficients are computed at the same saddle. Dividing cancels the common determinant and shell factor, not the evidence itself. The remaining $-\log Z(y)$ is precisely the state coordinate from Lemma 1.2. $\square$

1.4. Reachable chaotic beliefs and state reduction. Let $\mathfrak{R}_\varepsilon$ be the set of posterior currents reachable from the prepared initial family under admissible strategies and the declared observation channels. For $j \geq 2$ let $\mathfrak{h}_j(\Pi)$ be the connected j -particle hierarchy coordinate.

Lemma 1.5 (Posterior chaos on the reachable set). *There are $C, \rho < 1$ such that, uniformly over $\Pi \in \mathfrak{R}_\varepsilon$ and finite decision horizons,*

$$\|\mathfrak{h}_j(\Pi)\|_{\text{str}} \leq C^j j! \mu_\varepsilon^{1-j} + C \rho^j \varepsilon^\eta.$$

Proof. The prepared law has the cluster bound by B1. B2 graph propagation maps a connected j -cluster either to a connected kinetic cluster or to a recollision sector; the latter carries ε^η . The coarea update marks only finitely many coordinates and its evidence-lifted form does not divide the unnormalized connected current by a small Z . Normalize only after restricting to a finite evidence slice and then use the same ratio theorem. Induction over decision times gives the estimate. $\square$

Theorem 1.6 (Asymptotic sufficiency on strategy-reachable beliefs). *For bounded Lipschitz rewards and finite horizons, the value difference between two reachable beliefs with the same resolved kinetic state and evidence ledger is $o(1)$ uniformly over strategies. The limiting dynamic program is therefore posed on the resolved kinetic state. No equivalence is claimed for arbitrary correlated beliefs sharing only a one-particle mean.*

Proof. Expand one Bellman step in connected hierarchy coordinates. The zeroth and first coordinates are the resolved state and evidence. Every remaining term is bounded by the reward Lipschitz constant times the series in Lemma 1.5, which tends to zero uniformly. Backward induction over the finite number of decisions proves the result. $\square$

1.5. Risk-sensitive games and local statistics.

Theorem 1.7 (Risk-sensitive saddle and exact-centered LAN). *The evidence-filter kernel and quenched ratio theorem yield the finite-horizon risk-sensitive minimax recursion with measurable selectors. For a continuous prior with positive local density, the posterior has the LAN expansion centered at the exact finite-volume mean, and the local denominator is bounded below by the prior mass of a shrinking regular ball times the Gaussian coefficient.*

Proof. Feller continuity and compact containment give measurable minimax selectors. Theorem 1.4 passes each block payoff and transition kernel to the limit. Differentiate the exact finite log likelihood at its own mean; the linear score is exactly centered. Integrating the local quadratic likelihood over a ball on which the prior density has a positive minimum gives the explicit denominator. B3 supplies the limiting covariance on the quotient. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-eight dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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